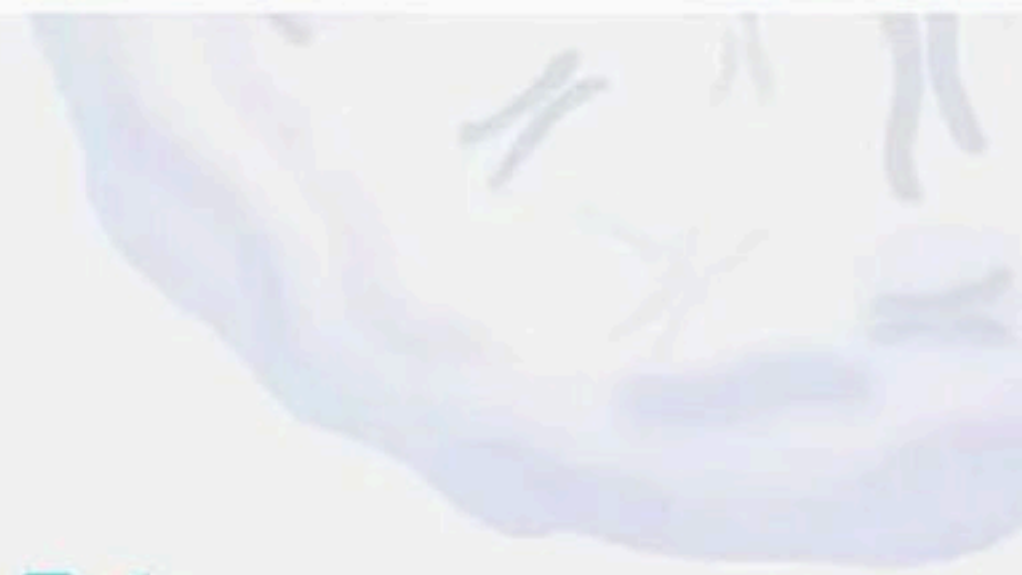
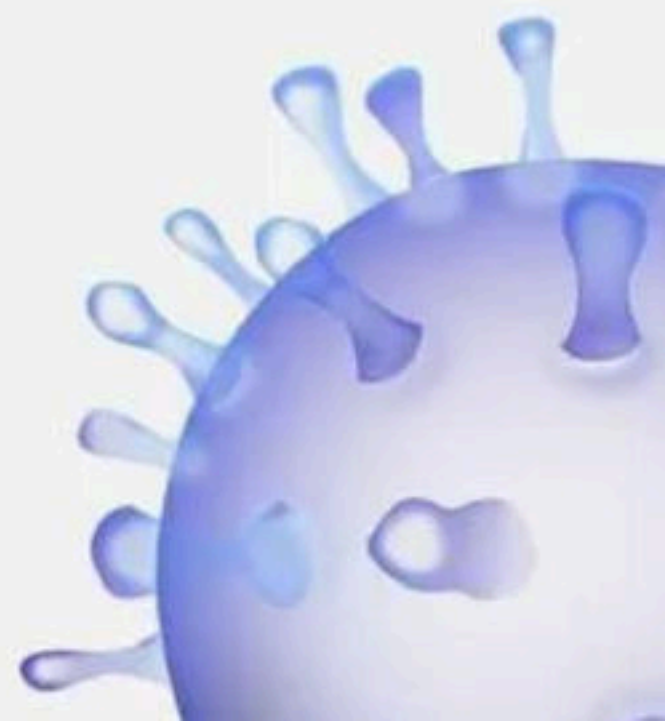


Cell: The Unit of Life - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Cell : The unit of life



BIOLOGYLIVE

1. Given below are two statements :

Statement I : Chromatin contain DNA and some basic protein called histones, some nonhistone protein and also RNA.

Statement-II : Material of nucleus stained by basic dyes was given name chromatin by *Robert Brown*.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



2. In the following How many matching are correct:

- | | | |
|-------------------|---|---------------------------|
| (A) Flagella | - | Axonems |
| (B) Mesosome | - | Analogous to mitochondria |
| (C) 70 S ribosome | - | Subunit 40 S and 30 S |
| (D) Lysosomes | - | Hydrolytic enzymes |

- (1) One
- (2) Two
- (3) Three
- (4) Four



3. Read the following statement and give correct answer regarding bacterial cell

(a) Glycocalyx is the outermost envelope in bacteria

(b) The glycocalyx could be a loose sheath called capsule

(c) The glycocalyx may be the thick and tough called slime layer

(1) a correct

(2) a, b correct

(3) a, c correct

(4) b correct

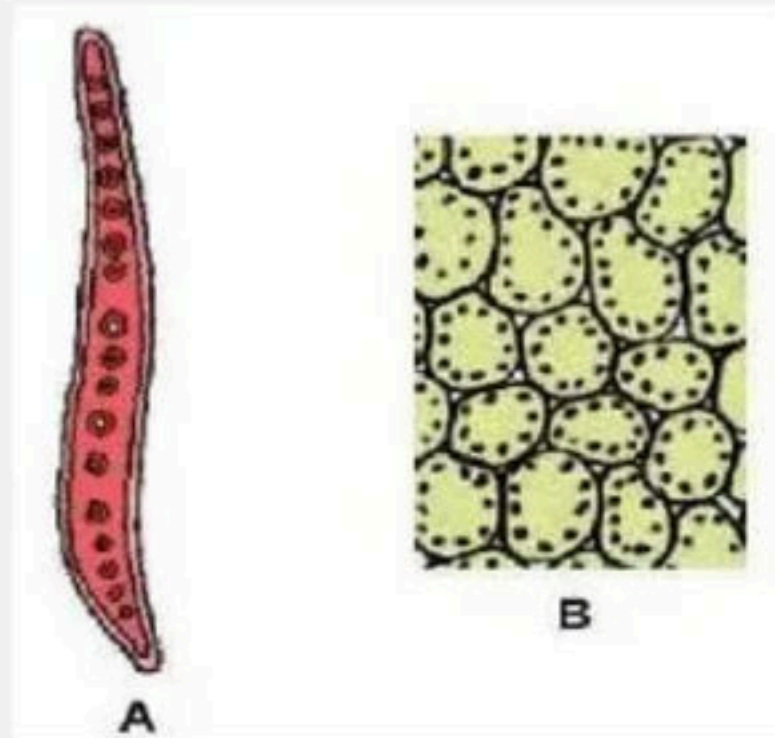


4. Which of the following is absent in Prokaryotes :

- (1) DNA
- (2) Mitochondria
- (3) RNA
- (4) Cell membrane



5. On the basis of given figures which of the following option is correct :



- (1) A – Vessel – Present in Xylem (2) B – Mesophyll cells – Round and oval
(3) A – Tracheid – Present in Xylem (4) More than one option correct



6. Which of the following is not includes in endomembrane system :

- (a) Golgi apparatus**
- (b) Endoplasmic Reticulum**
- (c) Mitochondria**
- (d) Lysosome**

- (1) a only
- (2) a and d only
- (3) c only
- (4) d only



7. Which of the following structure is an organelle within an organelle :

- (1) Mesosome
- (2) Mitochondria
- (3) Ribosome
- (4) Lysosome



8. Given below are two statements :

Statement I : Ribosomes are composed of RNA and proteins and are not surrounded by any membrane.

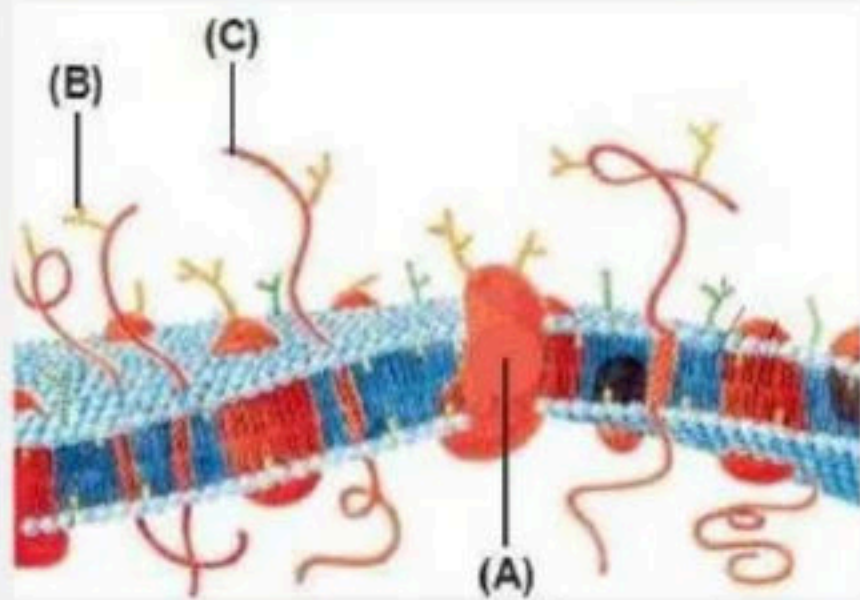
Statement-II : Eukaryotic ribosomes are 80 S and prokaryotic ribosomes are 70 S and are present in cytoplasm.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



9. In given diagram A, B and C represent



- (1) A-Integral protein, B-Peripheral Protein, C-Cholesterol
- (2) A-Peripheral Protein, B-Sugar, C-Cholesterol
- (3) A-Integral protein, B-Sugar, C- Peripheral Protein
- (4) A-Peripheral Protein, B-Cholesterol, C-Sugar



10. Which of the following does not differ in E.coli and *Spirogyra* :

- (1) Ribosome
- (2) Cell wall
- (3) Cell membrane
- (4) Mitochondria



11. In given structures how many structure are bound by double membrane.

Mitochondria, Ribosome, Vacuole, Lysosome, Chloroplast, centriole, Nucleus, Golgi apparatus

(1) 6

(2) 3

(3) 4

(4) 2



12. The main arena of various types of activities of a cell is:

- (1) Nucleus
- (2) Plasma membrane
- (3) Mitochondrion
- (4) Cytoplasm



13. Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP :

- (1) Mitochondrion
- (2) Lysosome
- (3) Ribosome
- (4) Chloroplast



14. Given below are two statements :

Statement I : Endoplasmic reticulum bearing ribosomes on their surface is called R.E.R.

Statement-II : R.E.R. is frequently observed in the cells actively involve in protein synthesis and secretion.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



15. Match the following scientists with their discovery and research year.

a) 1838

i. Matthias Schleiden

b) 1839

ii. Theodore Schwann

c) 1954

iii. Rudolf Virchow

d) 1855

iv. G. N. Ramachandran

(1) a-iii, b-iv, c-ii, d-i

(2) a-i, b-ii, c-iv, d-iii

(3) a-ii, b-iv, c-iii, d-i

(4) a-i, b-iv, c-iii, d-ii



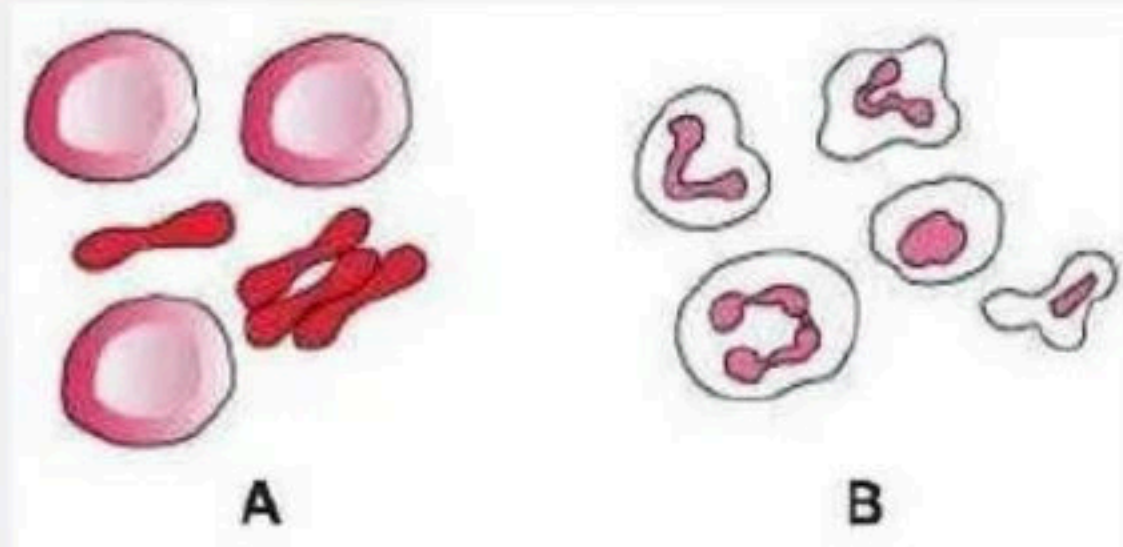
16. Which of the following cellular organelle is the site of formation of glycoproteins and glycolipids :

- (1) Peroxisomes
- (2) Mitochondria
- (3) Golgi apparatus
- (4) Chloroplast



17. Match the column I and column II**Column I****a) 1-2 μm** **b) 0.2 to 1.0 μm** **c) 3-5 μm** **d) 2-4 μm** **(1) a-iii, b-iv, c-ii, d-i****(3) a-ii, b-iv, c-iii, d-i****Column II****i) Typical bacteria****ii) Diameter of mitochondria****iii) Size of bacteria****iv) Chloroplast width****(2) a-i, b-ii, c-iv, d-iii****(4) a-i, b-ii, c-iii, d-iv**

18. According to given figure which of the following option is correct :



- (1) A – Red blood cells – Round and biconvex
- (2) B – White blood cells – Amoeboid
- (3) A – Red blood cells – about 3 μ m in diameter
- (4) All of these



19. Given below are two statements :

Statement I : Mitochondria is the largest cytoplasmic organelle of plants

Statement II : Ribosome is smallest membranous cell organelle in plant

Choose the correct answer from the options given below.

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct



20. The fluid nature of the membrane helps in :

- (1) Cell growth, secretion
- (2) Cell division
- (3) Formation of intercellular junctions
- (4) All of the above



21. The arrangement of lipid molecule in plasma

membrane found as:

- (1) Polar head towards the outer side and hydrophilic tails towards the inner side
- (2) Tails towards the outer side while polar head towards the inner side
- (3) Polar head towards the outer side and hydrophobic tails towards the inner side
- (4) Arranged alternatively with amino acids



22. In which among the following cell organelle, steroidal hormones are being synthesised in animal cells :

- (1) Smooth endoplasmic reticulum
- (2) Rough endoplasmic reticulum
- (3) Golgi complex
- (4) Vacuoles.



23. Match the following columns :

Column I

a. Typical bacteria

b. PPLO

c. Viruses

d. Eukaryotic cell

Column II

i. 0.02 - 0.2 μm

ii. 0.1 μm

iii. 10 - 20 μm

iv. 1 - 2 μm

(1) a-ii, b-iii, c-i, d-iv

(3) a-iv, b-ii, c-i, d-iii

(2) a-i, b-ii, c-iii, d-iv

(4) a-iii, b-i, c-ii, d-iv

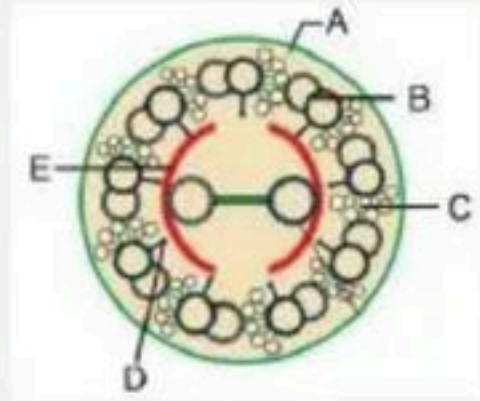


24. The cell wall of plants is made up of :

- (1) Hemicellulose + pectin
- (2) Hemicellulose + lipids + proteins
- (3) Hemicellulose + pectin + cellulose + protein
- (4) Hemicellulose + Chitin



25. Match A, B, C, D and E in diagram below from the list (i) to (v)



(i) Central sheath

(ii) Radial spoke

(iii) Plasma membrane

(iv) Interdoublet bridge

(v) Peripheral microtubules

(1) A-iii, B-v, C-iv, D-ii, E-i

(2) A-iii, B-iv, C-v, D-ii, E-i

(3) A-v, B-iii, C-iv, D-i, E-ii

(4) A-ii, B-v, C-iv, D-i, E-iii



26. Membrane-bound organelles are absent in:

- (1) *Streptococcus*
- (2) *Chlamydomonas*
- (3) *Plasmodium*
- (4) *Saccharmyces*



27. Polysome is formed by :

- (1) Ribosomes attached to each other in a linear arrangement
- (2) Several ribosomes attached to a single mRNA
- (3) One ribosomes attached to a single mRNA
- (4) A ribosome with several subunits



28. Middle lamella is composed mainly of :

- (1) Muramic acid
- (2) Calcium pectate
- (3) Phosphoglycerides
- (4) Hemicellulose



29. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Cells also vary greatly in their shapes.

Reason (R) : The shape of cell may vary with the function they perform.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



30. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : In Animal cells lipid like steroidal hormones are synthesised in a SER.

Reason (R) : Smooth endoplasmic reticulum is the major site for synthesis of protein.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



31. In given cell organelles how many organelles are bound by single membrane.

Mitochondria, Ribosome, Vacuole, Lysosome, Chloroplast, centriole, E.R, Golgi apparatus

(1) 6

(2) 5

(3) 4

(4) 2



32. Which of the following is incorrect statement about prokaryotic ribosome:

- (1) Ribosome is the site of protein sunthesis
- (2) 50s and 30s subunit unite to form 70s Ribsome
- (3) 60s and 40s subunit unite of form 80s Ribsome.
- (4) None of these



33. In prokaryotes like cyanobacteria, beside mesosome other membranous extensions into cytoplasm is/are :

- (1) Golgibody
- (2) Chromatophores
- (3) Ribosome
- (4) Mitochondria



34. The number of mitochondria per cell is variable, depending on :

- (1) Size of cells
- (2) Shape of cells
- (3) Physiological activity of cells
- (4) Type of genes present in mt DNA



35. In the following How many matching are correct:

- | | |
|------------------------|---|
| (A) Mesosome | - Found in Bacteria |
| (B) Fluid mosaic model | - Most correct model |
| (C) Gas vacuole | - Present in all prokaryote |
| (D) Mitochondria | - Site of aerobic and Anaerobic respiration |

- (1) One
- (2) Two
- (3) Three
- (4) Four



36. Different cells have different sizes. Arrange the following cells in an ascending order of their size. Choose the correct option among the followings.

I. Mycoplasma II. Ostrich eggs

III. Human RBC IV. Bacteria

1. I, IV, III, II

2. I, II, III, IV

3. II, I, III, IV

4. III, II, I, IV



37. Which of the following features is common to prokaryotes and many eukaryotes?

1. Chromatin material present
2. Cell wall present
3. Nuclear membrane present
4. Membrane-bound sub-cellular organelles present



38. Reserve material in prokaryotic cells are stored in the cytoplasm in the form of :

1. Pyrenoid
2. Parabodies bodies
3. Inclusion bodies which are bounded by a single membrane
4. Inclusion bodies which are not bounded by any membrane system



**39. How many of the given cell organelles are found only in prokaryotic cells?
[Vacuoles, Mesosomes, Ribosomes, Flagella, Cell wall, Mitochondria]**

1.1

2.2

3.3

4.4



40. Read the following statements and find the incorrect statement:

1. Prokaryotes are generally smaller and multiply more rapidly than eukaryotic cells.
2. Glycocalyx could be a loose sheath called a slime layer in some, while in others it is called the capsule.
3. The plasma membrane of prokaryotes is structurally dissimilar to that of eukaryotes.
4. Chromatophores are the membranous extensions into the cytoplasm of cyanobacteria which contain pigments.



41. Which of the following statement(s) about a bacterial cell is/are correct?

(i) Mesosome is formed by the extensions of the plasma membrane into the cell.

(ii) The pili are elongated tubular structures made up of a protein.

(iii) Flagellum is composed of filament, hook, and basal body.

(iv) Ribosomes are about 30 nm by 50 nm in size.

1.(i), (ii), and (iii)

2.(iii) and (iv)

3.(ii) and (iv)

4.None of the above



42. Bacterial cells have a chemically complex cell envelope. The cell envelope consists of a tightly bound three-layer structure, i.e., the:

1. Outermost cell wall followed by the plasma membrane and then the glycocalyx
2. Outermost glycocalyx followed by the plasma membrane and the cell wall
3. Outermost cell wall followed by the glycocalyx and then the plasma membrane
4. Outermost glycocalyx followed by the cell wall and then the plasma membrane



43. Which of the following is not true for a eukaryotic cell?

1. Cell wall is made up of peptidoglycans
2. It has 80S type of ribosome present in the cytoplasm
3. Mitochondria contain circular DNA
4. Membrane-bound organelles are present



44. Which of the following statement is not true for the plasma membrane?

1. It is present in both plant and animal cells
2. Lipid is present as a bilayer in it
3. Proteins are present integrated as well as loosely associated with the lipid bilayer
4. Carbohydrate is never found in it



45. The lipid component of the cell membrane mainly consists of:

1. Triglycerides
2. Phosphoglycerides
3. Glycolipids
4. Mono- and diglycerides



46. Phospholipid molecules of the cell membrane possess:

1. One polar head and one polar tail
2. One non-polar head and one non-polar tail
3. One polar head and two non-polar tails
4. One non-polar head and two non-polar tails



47. A student placed two cells in the same solution in two different containers. The observation is given in the table:

Container	Observation
1	Cell burst
2	Cell does not change its shape

Which structure maintains the shape of the cell present in container 2 and provides the most significant difference between the two cells?

- 1.Nucleus
- 2.Cell wall
- 3.Chloroplast
- 4.Cell membrane



48. If living cells similar to those found on Earth were discovered on another planet where there was no oxygen, which cell organelle would most probably be absent?

1. Cell membrane
2. Ribosome
3. Mitochondria
4. Chromosomes



49. What is common between chloroplasts, chromoplasts, and leucoplasts?

1. Presence of pigments
2. Presence of thylakoids and grana
3. Storage of starch, proteins, and lipids
4. Ability to multiply by a fission-like process



50. Consider the following statements:

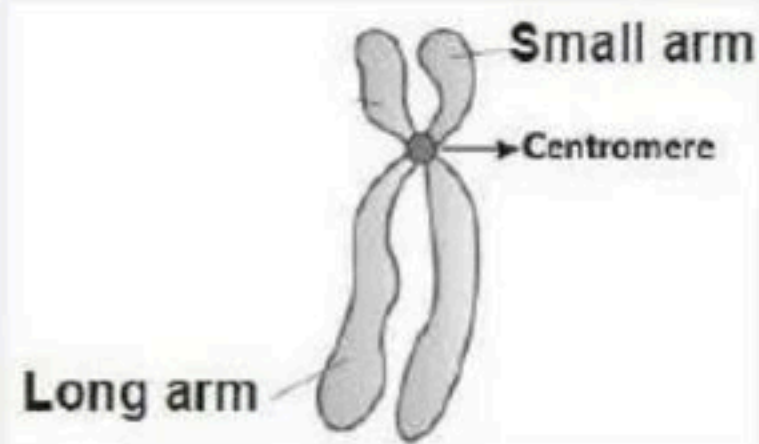
- A. In a prokaryotic cell, outgrowth of the plasma membrane into the cell is called polysome.**
- B. SER is the major site for glycoprotein synthesis.**
- C. RuBisCO is the most abundant protein of the biosphere.**
- D. Mitochondria, chloroplasts, and peroxisomes are not part of the endomembrane system.**

Which of the following statements are correct? :

- 1.C and D
- 2.C and B
- 3.B and A
- 4.A and D



51. In the given diagram the type of chromosomes is:



- (1) Telocentric
- (2) Metacentric
- (3) Acrocentric
- (4) Satalite chromosome



52. Analyse the following pairs and identify the correct option given below :

I. Chromoplasts	–	Contain pigments other than chlorophyll
II. Leucoplasts	–	Devoid of any pigments
III. Amyloplasts	–	Store oils and fats
IV. Aleuroplasts	–	Store carbohydrates
V. Elaioplasts	–	Store protein

(1) II and III

(2) III and IV

(3) IV and V

(4) I and II



53. Which among the statement is false :

- a. Basic unit of life is the cell in some living organism**
- b. Unicellular organisms are capable of independent existence**
- c. Unicellular organisms are capable of performing the essential function of life.**

Choose the option :

- | | |
|-------------|-------------|
| (1) a and b | (2) b and c |
| (3) Only a | (4) Only c |

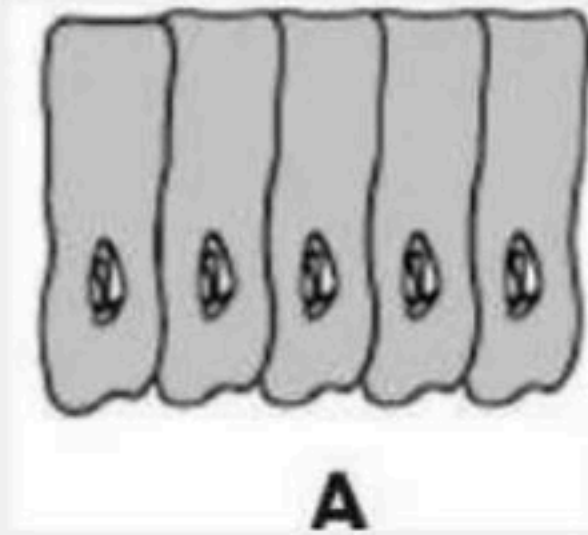


54. Flattened membranous sac forming the grana of chloroplasts

- (1) Thylakoids
- (2) Centrioles
- (3) Ribosomes
- (4) Lysosomes



55. According to given figure which of the following option is correct :



- (1) A – Columnar epithelial cells – Nuclei are located at the base of cell
- (2) A – Cuboidal Epithelium cells – Nuclei are located at the base of cell
- (3) A – Columnar epithelial cell – nuclei are absent
- (4) A – Cuboidal epithelial cell – nuclei are located at any place in cell



56. Given below are two statements :

Statement I : Centrosome contain two cylindrical structure called centrioles.

Statement II : Centrioles form the basal body of cilia or flagella, and spindle fibres.

Choose the correct answer from the options given below.

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both, Statement I and Statement II are correct



57. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Reserve material in Prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies.

Reason (R) : Inclusion bodies are bound by membrane system and lie free in cytoplasm.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is not correct but (R) is correct
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)



58. How many radial spokes are present in a cilium:

- (1) 18
- (2) 9
- (3) 27
- (4) 36



59. An elaborate network of filamentous proteinaceous structures present in the cytoplasm are called :

- (1) Cytoskeleton
- (2) Centrosome
- (3) Cilia
- (4) Flagella



60. Which among the following statement is incorrect:

- (1) Majority of the chloroplasts of the green plants are found in the mesophyll cells of the leaves.
- (2) The chloroplasts is double membrane bound the inner chloroplast membrane is relatively more permeable
- (3) Chlorophyll pigments are present in the thylakoids
- (4) Thylakoids are arranged in stacks like the piles of coins called grana



61. Given below are two statements :

Statement I : Centrioles are made up of nine evenly spaced peripheral fibril of tubulin protein.

Statement-II : Each peripheral fibril of centriol is doublet

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



62. Given below are two statements :

Statement I : Prokaryotic cells are generally smaller and multiply more rapidly than the eukaryotic cells.

Statement-II : Lysosome contain hydrolytic enzyme like lipases, proteases, carbohydrases optimally active at basic pH.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



63. How many matching are correct :

Organelles

Membrane, name/ character

A – Vacuoles

–

Double, Tonoplast

B – Chloroplast

–

Double, Inner more permeable

C – Mitochondria

–

Double, Inner form cristae

D – Centriole

–

Single, help in cell division

(1) 4

(2) 3

(3) 2

(4) 1



64. What is the percentage of lipid and protein in human erythrocyte respectively :

- (1) 50% , 40%
- (2) 52%, 40%
- (3) 40%, 52%
- (4) 80%, 40%.



65. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Endomembrane system include E.R, Golgi complex, lysosomes and vacuoles.

Reason (R) : Their functions are coordinated.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correction explanation of (A)



66. Gas vacuole are found in :

- (1) Red algae
- (2) Brown algae
- (3) Virus
- (4) Purple and green photosynthetic bacteria



67. Which one of the following cell organelle has its own DNA :

- (1) ER
- (2) Mitochondria
- (3) Dictyosome
- (4) Lysosome



68. Ribosomes are formed of :

- (1) rRNA + tRNA
- (2) mRNA + tRNA
- (3) rRNA + protein
- (4) mRNA + proteins



69. Most prokaryotic cell have a chemically complex cell envelop consisting of :

- (1) Glycocalyx
- (2) Cell wall
- (3) Plasma membrane
- (4) All of the above



70. What is the function of polysomes :

- (1) Translates DNA into proteins
- (2) Translates m-RNA into proteins
- (3) Translates r-RNA into carbohydrate
- (4) Translate m-RNA into DNA.



71. Which among the following is not present in animal cell :

- (1) Cell wall
- (2) Large central vacuole
- (3) Plastids
- (4) All of these



72. Identify figure and choose correct option for A and B :

(1) A – Actively Involved in protein synthesis and secretion

B – Major site for synthesis of lipid

(2) A – In Animal cell lipid like steroidal hormone synthesis

B – Major site for synthesis of lipid

(3) A – Major site for synthesis of lipid

B – Actively Involved in protein synthesis and secretion

(4) A – lipid like steroidal hormone synthesis

B – Major site for synthesis of protein



73. Many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called as:

- (1) Genomic DNA
- (2) snDNA
- (3) Plasmid
- (4) siDNA.



74. Match the following group I and II :

Group I

a. Amyloplast

b. Elaioplast

c. Aleuroplast

d. Chromoplast

e. Chloroplast

Group II

p. Carbohydrate

q. Xanthophyll

r. Protein

s. Fat

t. Chlorophyll

(1) a-p, b-r, c-s, d-q, e-t

(3) a-s, b-p, c-r, d-q, e-t

(2) a-p, b-s, c-r, d-q, e-t

(4) a-r, b-s, c-p, d-q, e-t



75. Arrange the following cells in an ascending order of their sizes

(I) Mycoplasma

(II) Ostrich egg

(III) Human RBC

(IV) Bacteria

(1) I, II, III, IV

(2) I, IV, III, II

(3) II, IV, I, III

(4) IV, III, II, I



76. The chromosomes in which centromere is situated at terminal :

- (1) Telocentric
- (2) Sub-metacentric
- (3) Metacentric
- (4) Acrocentric



77. Which of the following is not true regarding cell membrane

- (a) depending on ease of extraction peripheral and integral two types of protein are present in cell emmbrane**
- (b) peripheral protein lie on surface of membrane**
- (c) integral protein are partially or totally buried in membrane**
- (d) ratio of protein and lipids remains same in different type of cells**

- (1) a and d not true**
- (2) d not true**
- (3) all are true**
- (4) None of these**



78. If you remove the fimbriae from the bacterial cell, which of the following would you expect to happen :

- (1) The bacteria could not longer swim
- (2) The bacteria would not adhere to the host tissue
- (3) Transportation of molecules across the membrane would stop
- (4) The shape of bacteria would change



79. Read the given statements regarding a cell organelle.

- I. It contains water, sap, excretory products and other unwanted materials**
- II. It is bounded by a single membrane called tonoplast**
- III. In plant cells, it can occupy upto 90% of cellular volume.**

The above features are attributed to :

- | | |
|----------------|-------------------|
| (1) Lysosome | (2) Vacuole |
| (3) Peroxisome | (4) Mitochondrion |

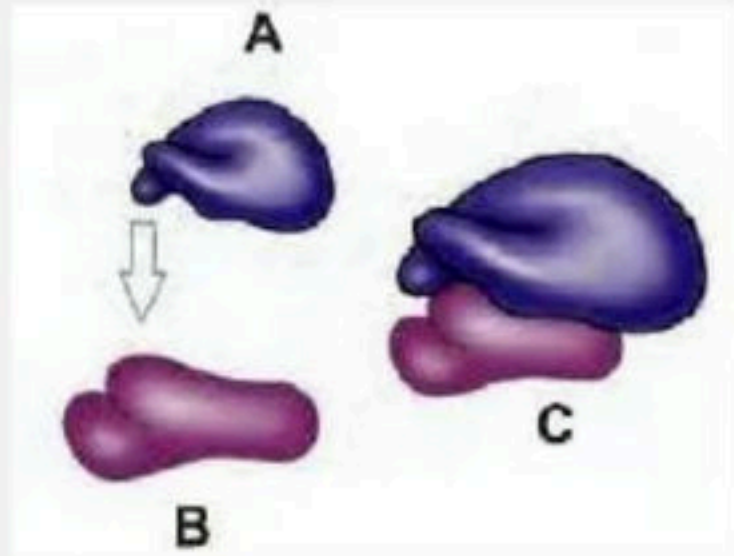


80. The core of a cilium or flagellum composed of microtubules and their associated proteins is called:

- (1) Blepharoplast
- (2) Tubulin
- (3) Microfilament
- (4) Both 1 and 2



81. Recognise the given figure and choose correct option:



- (1) A – 50S, B - 30S, C – 80S
- (2) A – 50S, B - 30S, C – 70S
- (3) A – 30S, B - 50S, C – 70S
- (4) A – 40S, B - 60S, C – 80S



82. Material of nucleus stained by

- (1) Acidic dye
- (2) Basic dye
- (3) Neutral dye
- (4) None of these



83. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as :

- (1) Telocentric
- (2) Sub-metacentric
- (3) Acrocentric
- (4) Metacentric



84. Which structure perform the function of mitochondria in bacteria :

- (1) Nucleoid
- (2) Ribosomes
- (3) Cell wall
- (4) Mesosomes



85. Satellite means:

- (1) Terminal part of the chromosome beyond secondary constriction.
- (2) Terminal part of the chromosome beyond primary constriction.
- (3) Terminal part of the chromosome beyond tertiary constriction.
- (4) None of the above



86. A cell organelle 'X' is divided into two types on the basis of a cell organelle 'Y', that helps in protein synthesis. Identify 'X' and 'Y' respectively.

- (1) Golgi complex and ribosome
- (2) SER and mitochondria
- (3) ER and ribosome
- (4) Lysosome and ER



87. Which of the following statement is true for a secretory cell?

- (1) Golgi apparatus is absent
- (2) Rough Endoplasmic Reticulum (RER) is easily observed in the cell.
- (3) Only Smooth Endoplasmic Reticulum (SER) is present
- (4) Secretory granules are formed in the nucleus



88. The organelle 'X' is associated with the detoxification of drugs. It involves the muscle contraction by the release and uptake of Ca^{2+} ions. Identify 'X'.

- (1) Golgi complex
- (2) RER
- (3) SER
- (4) Free ribosomes



89. Read the statements given below with regard to the functions performed by the Golgi apparatus:

- (i) Transport and chemically modify the materials contained within it.**
- (ii) Performs the function of packaging materials.**
- (iii) Important site of formation of glycoproteins and glycolipids.**

Which of the following is the correct answer?

- (1) (i) is wrong but (ii) and (iii) are correct
- (2) (ii) is wrong but (i) and (iii) are correct
- (3) (ii) and (iii) are wrong but (i) is correct
- (4) All are correct



▲ 3 • Asked by Fatima

Q39 mai eu mai bhi hota h 70s hota h

90. How many of the following properties are related to lysosomes?

Membrane-bound vesicular structure, hydrolases, basic pH, formed in the RER, capable of digesting nucleic acids.

- (1) Two
- (2) Three
- (3) Four
- (4) Five



91. Choose the incorrect match:

- (1) Nucleus — RNA
- (2) Lysosome — Protein synthesis
- (3) Mitochondria — Respiration
- (4) Cytoskeleton — Microtubules



92. Lysosomes are produced by:

- (1) Golgi complex
- (2) Mitochondria
- (3) Endoplasmic reticulum
- (4) Leucoplasts



93. Hydrolytic enzymes like lipases, proteases, and carbohydrases are found in:

- (1) Lysosome
- (2) Golgi Bodies
- (3) ER
- (4) Mitochondrion



94. Select one which is not true for ribosome:

- (1) Made up of two sub-units
- (2) Form polysome
- (3) May attach to mRNA
- (4) Have no role in protein synthesis



95. If A = 70S ribosomes, B = 80S ribosomes, C = single circular DNA, D = RNA molecules, E = double helical linear DNA, F = thylakoids, and G = stromal lamellae, which of the following set is found in mitochondria?

- (1) B, D, E, F
- (2) D, E, F, G
- (3) A, D, E
- (4) A, C, D



96. Which of the following pairs are correctly matched?

- A. Microtubules – Structural components of cilia**
- B. Centrioles – Store hydrolytic enzymes**
- C. Amyloplasts – Store oil, protein, and starch in plants**

(1) A, B, and C

(2) A and B

(3) A only

(4) A and C



97. Given below are some characters of a cell organelle. Identify the correct organelle which shows all the characters described above:

- (i) It is a membrane-bound space found in the cytoplasm.**
- (ii) It is bound by a single membrane called tonoplast.**
- (iii) It contains water, sap, excretory products, and other materials not useful to the cell.**
- (iv) It has a higher concentration of sap than the cytoplasm.**

- (1) Golgi apparatus
- (2) Lysosomes
- (3) Endoplasmic reticulum
- (4) Vacuoles



98. Which of the following statement is correct regarding vacuole?

- (1) It is membrane-bound and contains storage proteins and lipids.
- (2) It is membrane-bound and contains water and excretory substances.
- (3) It lacks membrane and contains air.
- (4) It lacks membrane and contains water and excretory substances.



99. Fill in the blanks:

1. In plant cells, the vacuoles can occupy up to ...A... of the volume of the cell.
2. In plants, the tonoplast facilitates the transport of a number of ions and other materials ...B... in concentration gradient into the vacuole, hence their concentration is significantly ...C... in the vacuole than in the cytoplasm.
3. In Amoeba, the ...D... vacuole is important for excretion.
4. In many cells, as in protists, ...E... vacuoles are formed by engulfing the food particles.

- (1) A - 70%, B - along, C - higher, D - food, E - contractile
- (2) A - 70 to 90%, B - against, C - higher, D - contractile, E - food
- (3) A - 90%, B - against, C - lower, D - contractile, E - inclusion
- (4) A - 90%, B - against, C - higher, D - contractile, E - food



100. Which of the following is not a function of cytoskeleton in a cell?

- (1) Protein synthesis
- (2) Maintenance of cell shape and structure
- (3) Support of the organelle
- (4) Cell motility



THANK YOU



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